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30/01/2026

M1 Actuariat
ISFA

IMPACT OF EXCESS-OF-LOSS REINSURANCE ON CATASTROPHE RISK IN NON-LIFE INSURANCE

Contents

1 Abstract

1.1 Abstract

This project investigates the impact of excess-of-loss reinsurance on the loss distribution of a non-life insurance portfolio exposed to catastrophic risks. First, aggregate losses are modeled using a frequency-severity framework, combining a Poisson distribution for claim frequency with a heavy-tailed distribution for claim severity.

Monte Carlo simulations are then performed to estimate the annual loss distribution and assess risk using standard measures such as Value-at-Risk *(VaR)* and Tail Value-at-Risk *(TVaR)*. In a second step, an excess-of-loss reinsurance treaty is introduced, and its effect on the loss distribution is analyzed.

The results show that reinsurance significantly reduces extreme losses and mitigates tail risk, with a particularly strong impact on TVaR. This study highlights the key role of reinsurance in managing catastrophic risk exposure for insurance companies.

Following sections items are clickable

Our line of reasoning can be followed via the GitMind link provided below :

T.E.R gitmind link

1.2 Forewords

We refere to results and demonstrations writted in auxiliar papers. In a first time if the github link doesent show the paper, just refresh the page.

If the github link still doesent work, you can contact us in a second time at : mhreinette@gmail.com. A other version of this paper is availible in a black theme on this present link.

All the ressources and notebook are availible on this github repository. Different colors is used to caracterise : définitions , theorem , important results , crit ics .

Part I

Frequency-Severity Model

2 T.E.R.

3 Frequency

We are less concerned here about the physical reality of the Poisson model than with its effectiveness. A Poisson model can be more or less difficult to calibrate, depending on the model assumptions and the historical data provided.
We do not believe in the reality of a homogeneous Poisson model (which seems even more improbable for modeling flood damage claims).

Following the principle of parsimony, we must prefer simplicity over complexity, provided that the former is sufficiently robust and predictive.
The difficulty of such models lies not in their algorithmic complexity, but rather in the verification of their underlying assumptions.

We will therefore proceed by increasing the level of complexity regarding our intensity assumptions.
Consequently, we must design point estimation methods.

3.1 SUB CASE: CONSTANT INTENSITY

○ Filtered Space : $(\Omega,\ \mathcal{A},\ \mathcal{F} = (\mathcal{F}_t)_{t\in T}\ ,\ \mathbb{P})$

Let : $(\Omega,\ \mathcal{A},\ \mathcal{F} = (\mathcal{F}_t)_{t\in T}\ ,\ \mathbb{P})$, define by :
— $(\Omega,\ \mathcal{A},\ \mathbb{P})$: a probabilised space
— $\mathcal{F} = (\mathcal{F}_t)_{t\in T}$: A growing succession of σ-algrebra of Ω

definitions and notations

○ Homogeneous poisson process : $R^\lambda = (R_t^\lambda)_{t\in T}$

Let : $R^\lambda = (R_t^\lambda)_{t\in T}$, define by :
— $(\Omega,\ \mathcal{A},\ \mathcal{F} = (\mathcal{F}_t)_{t\in T}\ ,\ \mathbb{P})$: a filtered space
— $\lambda \in \mathbb{R}_+^*$, the intensity of this poisson process
— R^λ, $\mathcal{F}$-adpted process

definitions and notations

Idea

Recall :

○ Laplace Transformation : $L_X(t)$

Let : $L_X(t)$, define by :
— X, a random variable

$$\forall t \in \mathbb{R}, L_X(t) = E[exp(tX)]$$

definitions and notations

◇ theorem

Hypothesys :
— X, a random variable
— Y, a random variable
— $O \subseteq \mathbb{R}$, a open, where $0 \in O$
— $L_X \approx L_Y$, on O

Conclusions :
— $X \sim Y$

theorem

$U \sim \mathcal{P}(\beta)$, where $\beta \in \mathbb{R}^+ \iff \forall t \in \mathbb{R},\ L_U(t) = exp(\beta(e^t - 1))$

$\iff (t_a, t_b)$, where $0 < t_a < t_b$: $\forall t\mathbb{R} \in L_{\alpha(R^\lambda_{t_b} - R^\lambda_{t_a})}(t) = exp(\lambda(t_b - t_a)(e^{t\alpha} - 1))$

We want to fund $\alpha(t_a, t_b)$, such that : $\forall t\mathbb{R} \in L_{\alpha(R^\lambda_{t_b} - R^\lambda_{t_a})}(t) = exp(\lambda(t_b - t_a)(e^{t\alpha} - 1)) = exp(\lambda(e^t - 1))$

exp is a *bijective*, increasing function, so :
$exp(\lambda(t_b - t_a)(e^{t\alpha} - 1)) = exp(\lambda(e^t - 1)) \iff \lambda(t_b - t_a)(e^{t\alpha} - 1) = \lambda(e^t - 1)$

$$\iff (t_b - t_a)(e^{t\alpha} - 1) = (e^t - 1) \iff e^{t\alpha} = 1 + \frac{e^t - 1}{t_b - t_a}$$

$$\iff (t_b - t_a)(e^{t\alpha} - 1) = (e^t - 1) \iff e^{t\alpha} = 1 + \frac{e^t - 1}{t_b - t_a}$$

$$\iff t\alpha(t) = \underbrace{ln(1 + \frac{e^t - 1}{t_b - t_a})}_{=_{t\to 0} \frac{e^t - 1}{t_b - t_a} =_{t \to 0} \frac{t}{t_b - t_a}}$$

$$\iff \alpha \approx \frac{1}{t_b - t_a}, \text{ on }]0,1[, \text{ a open.}$$

◇ Lemma

Hypothesys :
— $R^\lambda = (R^\lambda_t)_{t\in T}$, a homogeneous Poisson process
— $H \in \mathbb{R}^+$
— $\forall n \in \mathbb{N}^*, \forall(t_1, t_2, ..., t_n)$, a strictly growing succession of $[0, H]^n$

Conclusions :
— $(\frac{R^\lambda_{t_1}}{t_1}, \frac{R^\lambda_{t_2} - R^\lambda_{t_1}}{t_2 - t_1}, ..., \frac{R^\lambda_{t_n} - R^\lambda_{t_{n-1}}}{t_n - t_{n-1}})$, i.i.d. random variables
— $\forall i \in [|2, n|]$, $R^\lambda_{t_i} - R^\lambda_{t_{i-1}} \sim \mathcal{P}(\lambda(t_i - t_{i-1}))$, and $R^\lambda_{t_1} \sim \mathcal{P}(\lambda(t_1))$

theorem

Demonstration

$R^\lambda = (R^\lambda_t)_{t\in T}$ is a independent increments process. $t_1 < t_2 < ... < t_n$, so $(R^\lambda_{t_1}, R^\lambda_{t_2} - R^\lambda_{t_1}, ..., R^\lambda_{t_n} - R^\lambda_{t_{n-1}})$, is i. random variables.

$$\iff (\frac{R^\lambda_{t_1}}{t_1}, \frac{R^\lambda_{t_2} - R^\lambda_{t_1}}{t_2 - t_1}, ..., \frac{R^\lambda_{t_n} - R^\lambda_{t_{n-1}}}{t_n - t_{n-1}}), \text{ i. random variables}$$

$\forall i \in [|2, n|]$, $R^\lambda_{t_i} - R^\lambda_{t_{i-1}} \sim \mathcal{P}(\lambda(t_i - t_{i-1}))$, and $R^\lambda_{t_1} \sim \mathcal{P}(\lambda(t_1))$, by the Poisson identity.

$$\iff (\frac{R^\lambda_{t_1}}{t_1}, \frac{R^\lambda_{t_2} - R^\lambda_{t_1}}{t_2 - t_1}, ..., \frac{R^\lambda_{t_n} - R^\lambda_{t_{n-1}}}{t_n - t_{n-1}}), \text{ i.i.d. random variables.}$$

○ Estimator : ${}^\lambda R^*_n$

Let : ${}^\lambda R^*_n$, define by :
— $R^\lambda = (R^\lambda_t)_{t\in T}$, a homogeneous Poisson process
— $t_0 = 0$ ($R_0 = 0$ a.s)
— $H \in \mathbb{R}^+$
— $\forall n \in \mathbb{N}^*, \forall(t_1, t_2, ..., t_n)$, a strictly growing succession of $[0, H]^n$

$${}^\lambda R^*_n((t_1, t_2, ..., t_n)) = \frac{1}{n}\sum_{i=0}^{n-1} \frac{R^\lambda_{t_{i+1}} - R^\lambda_{t_i}}{t_{i+1} - t_i}$$

— Henceforth, we will use the notation ${}^\lambda R^*_n$ to denote ${}^\lambda R^*_n$ of arbitrary samples $(t_1, t_2, ..., t_n)$

definitions and notations

◇ theorem

Hypothesys :
— $R^{\lambda} = (R_t^{\lambda})_{t \in T}$, a homogeneous Poisson process

Conclusions :
— $\boxed{{}^{\lambda}R_n^* \xrightarrow{\mathbb{P}} \lambda}$
— $\boxed{E[\ {}^{\lambda}R_n^*] = \lambda}$

theorem

Demonstration

For the first assumption $\forall i \in [|0, n-1|]$, $\frac{R_{t_{i+1}}^{\lambda} - R_{t_i}^{\lambda}}{t_{i+1} - t_i}$, is identically distributed, so we can applied the **Large Number Theorem**.

$$E[\ {}^{\lambda}R_n^*((t_1, t_2, ..., t_n))] = E[\frac{1}{n}\sum_{i=0}^{n-1} \frac{R_{t_{i+1}}^{\lambda} - R_{t_i}^{\lambda}}{t_{i+1} - t_i}] = \frac{1}{n}\sum_{i=0}^{n-1} \frac{1}{t_{i+1} - t_i} E[R_{t_{i+1}}^{\lambda} - R_{t_i}^{\lambda}]$$

$$= \frac{1}{n}\sum_{i=0}^{n-1} \frac{1}{t_{i+1} - t_i} \lambda(t_{i+1} - t_i) = \frac{1}{n}\sum_{i=0}^{n-1} \lambda = \lambda$$

3.2 SUB CASE: LOCAL INTENSITY

Inhomogeneous Poisson process, is a little bit more abstract and difficult to simulate.
However the true difficulty reside in the *model calibration*.
For a homogeneous poison process ; the intensity is constant, a debatable assumption (especially in Cat-Nat simulation).

To simplify the problem, we will make several stability assumptions regarding our intensity function, allowing us to treat it as locally constant.

○ Inhomogeneous Poisson process : $R^{|f|} = (R_t^{|f|})_{t \in T}$

Let : $R^{|f|} = (R_t^{|f|})_{t \in T}$, define by :
— $f \in L^2(\mathbb{R} \to \mathbb{R}_+^*)$

$$\forall t \in \mathbb{R}^+, R_t^{|f|} \sim \mathcal{P}(\int_0^t f(x)dx)$$

definitions and notations

To estimate f, we will have no choice but to discretize our measurements. We rely on the principle of local intensity estimation (assuming that the intensity is constant in the neighborhood of chosen points x_i : $f = \lambda$).

Since almost all the information is contained in the jumps of our process, we need a neighborhood that is neither too small, to ensure we have sufficiently distinct measurements, nor too large, so that our hypothesis remains valid.

◦ Local Model : $M^{(\flat,\ell,m)}$

Let : $M^{(\flat,\ell,m)}$, define by :
— $\flat \in \mathbb{R}_+^*$: *Bias-Variance/Bandwidth*
— $\ell \in \mathbb{R}^+$: *Horizon*
— $m \in \mathbb{N}^*$: *local monte carlo frequency*

definitions and notations

We will collect the values R_{t_i} from m points normally distributed around t with a variance of $\flat$.

◦ Global Local Model : ${}^kM^{(\flat,\ell,m)}$

Let : ${}^kM^{(\flat,\ell,m)}$, define by :
— $M^{(\flat,\ell,m)}$, a local model
— $k \in \mathbb{N}^*$: *segmentation* (we will assume we have k local intensities)
— $0 < t_1 < t_2, ..., t_k$
— $f \in C^0(\mathbb{R} \to \mathbb{R}_+^*)$, borned
— $\forall x \in \mathbb{R}$, $g : \mathbb{R} \to [0, \ell[$
— $\forall i \in [|1, k|]$, ${}^{\lambda_i}R_m^* = {}^{\lambda_i}R_m^*((g(t_1 + N(0, \flat^2)), g(t_1 + N(0, \flat^2)), ..., g(t_1 + N(0, \flat^2)))_{ordonated})$, a estimation of :

$$\lambda_i \approx f(t_i)$$

— $Est((t_1, t_2, ..., t_k)) = ({}^{\lambda_1}R_m^*, ..., {}^{\lambda_k}R_m^*)$, our estimations of the local intensity.

definitions and notations

This method is somewhat imperfect. As previously mentioned, it assumes a degree of stability in our local deterministic intensity; it is therefore up to us to choose the t_i segmentation to highlight periods of high variation. Furthermore, the method is quite sensitive to the Local Model's hyperparameters. It is also worth noting that the results are more relevant for middle local intensities than for the beginning and final ones.

Demonstration

We choose 2 points : $0 < t_a < t_b$, where t_b is close to t_a.

$$R_{t_b} - R_{t_a} \sim \mathcal{P}(\int_0^{t_b} f(x)dx - \int_0^{t_a} f(x)dx) \sim \mathcal{P}(\int_{t_a}^{t_b} f(x)dx)$$

But we say in the neighborhood of t_a, f is praticly constant.
So $\exists c \in \mathbb{R}_+^*$, such that : $f_{|neighborhood} \approx c = f(t_a) \iff$

$$R_{t_b} - R_{t_a} \sim \mathcal{P}(\int_{t_a}^{t_b} f(t_a)dx) \sim \mathcal{P}(f(t_a)(t_b - t_a))$$

We have a point t_i.
We can generate : $(X_1, ..., X_m) \sim (g(t_i + N(0, \flat^2), ..., g(t_i + N(0, \flat^2))$, a increasing succession of positive points close to t_i.

At this point we can refer to out last result on the constant intensity and :

$$^{f(t_i)}R_m^*(X_1, ..., X_m) \xrightarrow{\mathbb{P}} f(t_i) = \lambda_i$$

$$E[\ ^{f(t_i)}R_m^*(X_1, ..., X_m)] = f(t_i) = \lambda_i$$

Our model may be interesting in itself; however, we built it to serve as a stepping stone toward a more general model. The local model allows us to get an idea of the overall shape of our intensity.

(EMPIRICAL REVIEW): Local Intensity Model

Forewords

Foreword 01 :
In this section, we will only use synthetic data to allow for comparison. We need to quantify our models' performance against known data (the simplest method being to generate them ourselves).

Foreword 02 :
In this section, we will not discuss our implementation methods (which, while not fully optimized, are not naive implementations). These will be freely discussed in the Jupyter notebook below (within code comments or markdown cells) :

Second notebook link

Foreword 03 :
For the purpose of our experiments, we have used three different notebooks:

— The first focuses solely on the proper implementation of our models (primarily code-heavy).

— The second covers empirical tests conducted using synthetic data.

— The last one utilizes our historical data to build our final model.

We are fully aware that estimating an intensity function from a single trajectory is ambitious at best, and in this sense, our initial results might have appeared fragile.

This section is significant because, despite our more or less solid theoretical results, implementation is not necessarily straightforward. It is neither trivial nor a direct corollary of our previous findings.

The noise

We can establish an initial observation: our local intensity model is extremely sensitive to hyperparameters.

The most obvious example lies in the « *Bandwidth* » parameter : $\flat$, which has a significant influence on the volatility of our estimates.

This appears to have a slight influence on both the bias and variance of our estimator. However, lacking more powerful models, we cannot determine whether these results are due to pure chance or the influence of the bandwidth.

That said, its importance should be put into perspective, as it implicitly depends on the choice of the normalizing function g.

Where the bandwidth clearly stands out is in how it reduces the extreme values of our local estimators (decreasing efficiency on inhomogeneous processes while increasing it on homogeneous ones); it smooths out the noise while maintaining a remarkably stable variance.

However, it should be noted that (due to our choice of the function g), the time complexity of our model is an increasing function of the bandwidth.

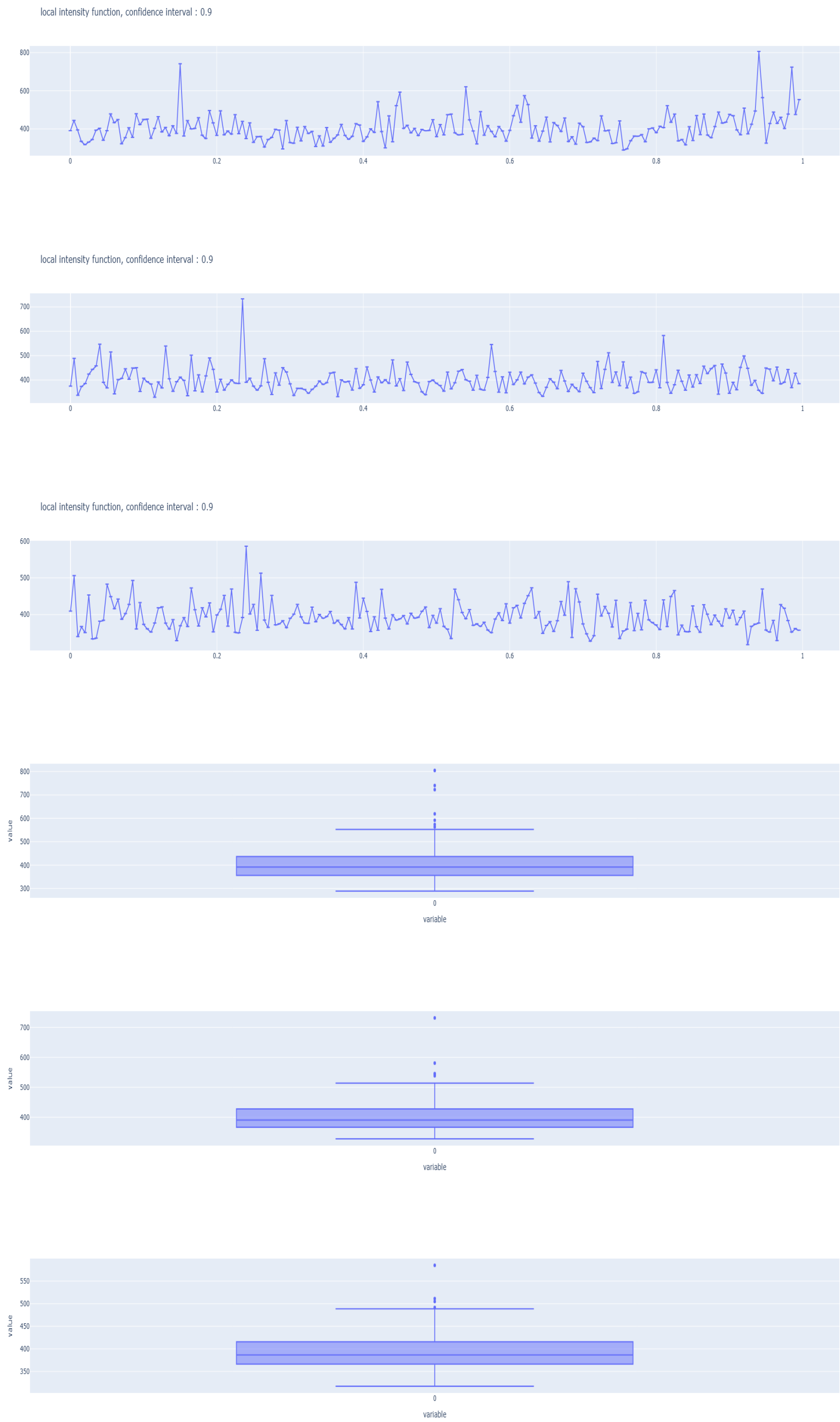


Figure 1: homogeneous process

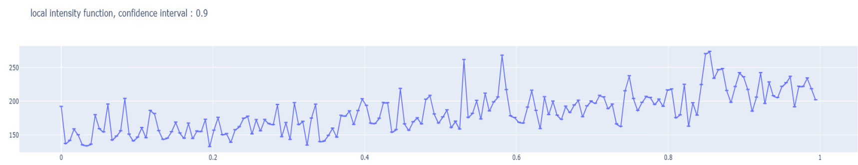
Figure 2: inhomogeneous process

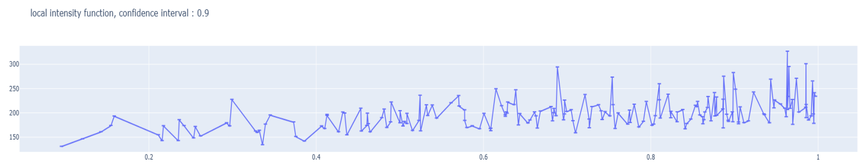
The m parameter (*the local monte carlo frequency*), for its part, affects the reliability of our estimators (although this effect is asymptotically smoothed out).

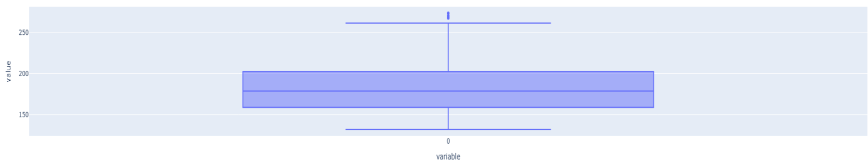
Secondly, we must debunk a persistent intuition: we had previously considered centering our estimators around the jumps of our trajectory (choosing t_i as the jump point T_i of our process R). We believed this strategy to be optimal, assuming that the majority of the information was contained within these jumps.
This last strategy caused the severity of our outliers of our estimates to explode in certain configurations, drowning out general trends in noise. These results were alarming at first, until we considered changing our sampling strategy. Instead of selecting our estimators based on the jumps in our trajectory, we used a simpler strategy by subdividing our interval $[0, \ell]$ into regular subdivisions, independent of our jumps, to place the estimators.

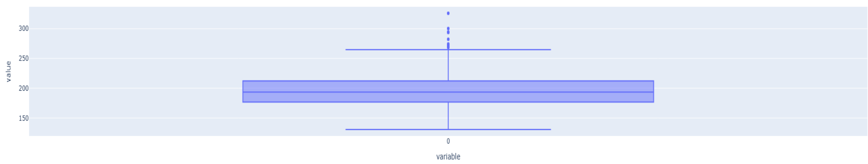
Figure 3: homogeneous process ($\lambda = 400$)

Noting that ♭ and the change in sampling had opposite effects on outliers (and to a much lesser extent on variance), the idea of combining them naturally arose to see if their effects would cancel each other out.









While the result may initially seem disappointing (given a linear intensity function f with a coefficient of 400), we have nonetheless allowed ourselves the following bold and purely speculative comparison:

> This change of perspective/measure drowns our signal (or general trend) in noise, or even eliminates it entirely, while leaving our volatility almost unchanged. This is reminiscent of changes of measure in stochastic calculus, which transform an Ito process into a martingale by eliminating its drift without modifying its volatility.

$$dX_t = \underbrace{\mu(t, X_t)dt}_{=0 \text{ under } \mathbb{Q}} + \sigma(t, X_t)dW_t$$

Other attempts were made to improve our model (which can be found in the second notebook). While we will not discuss them in detail in this section, we can note that the regular subdivision model shows significant sensitivity to the 'number of subdivisions' hyperparameter.

Noise reduction attempts were also made using Fast Fourier Transform (FFT); however, this ultimately adds more hyperparameters to a problem that already has enough.

Furthermore, adding an FFT filter would introduce overfitting issues (due to our hyperparameters). We must therefore act in accordance with the principle of parsimony.

In the interest of intellectual honesty, it should be noted that these trends only became apparent for sufficiently regular common functions. We are willing to admit that in such cases, an FFT filter could prove interesting (disregarding overfitting issues).

3.3 Confidence interval for local intensity

◇ Lemma

Hypothesys :
— $\forall n \in \mathbb{N}^*$

Conclusions :

$$\mathbb{P}(\ {}^{\lambda}R_n^* - Z_{1-\frac{\alpha}{2}} \sqrt{\frac{{}^{\lambda}R_n^*}{n}} \leq \lambda \leq \ {}^{\lambda}R_n^* + Z_{1-\frac{\alpha}{2}} \sqrt{\frac{{}^{\lambda}R_n^*}{n}}) \underset{n \to +\infty}{\longrightarrow} 1 - \alpha$$

theorem

Demonstration

$$ {}^{\lambda}R_n^* \xrightarrow[n]{\mathbb{P}} \lambda \iff \sqrt{\frac{\lambda}{{}^{\lambda}R_n^*}} \xrightarrow[n]{\mathbb{P}} 1$$

$$\sqrt{n}\frac{{}^{\lambda}R_n^* - \lambda}{\sqrt{\lambda}} \xrightarrow[n]{\mathcal{L}} N(0,1), \ \textit{central theorem limit}$$

With this 2 results, we can now use the *Slutsky theorem* :

$$\sqrt{n}\frac{{}^{\lambda}R_n^* - \lambda}{\sqrt{\lambda}}\sqrt{\frac{\lambda}{{}^{\lambda}R_n^*}} = \sqrt{n}\frac{{}^{\lambda}R_n^* - \lambda}{\sqrt{{}^{\lambda}R_n^*}} \xrightarrow[n]{\mathcal{L}} N(0,1) \iff$$

$$\mathbb{P}(\ {}^{\lambda}R_n^* - Z_{1-\frac{\alpha}{2}} \sqrt{\frac{{}^{\lambda}R_n^*}{n}} \leq \lambda \leq \ {}^{\lambda}R_n^* + Z_{1-\frac{\alpha}{2}} \sqrt{\frac{{}^{\lambda}R_n^*}{n}}) \underset{n \to +\infty}{\longrightarrow} 1 - \alpha$$

3.4 GENERAL CASE: CONTINUOUS INTENSITY

We maintain the same inhomogeneous Poisson process. However, instead of focusing on the process itself, the occurrence times will serve as our subject of study.

We will assume here that f is known (or at least that we suspect f to be the intensity at the origin of our Poisson process trajectory). Note: f is always deterministic.

◦ Increasing in time : X_i

Let : X_i, define by :
— $T_0 = 0$
— $(T_1, ..., T_n) \in \mathbb{R}_+^n$: the times of jumps of our process
— $\forall i \in [|1, n|]$, $X_i = T_i - T_{i-1}$

definitions and notations

◦ Integration function : ∇_f

Let : ∇_f, define by :
— $\forall t \in \mathbb{R}_+$,

$$\nabla_f(t) = \int_0^t f(x)dx$$

definitions and notations

Let us review some fundamental concepts:

Recalls

⋄ Lemma

Hypothesys :
— $(T_1, ..., T_n) \in \mathbb{R}_+^n$: the times of jumps of a homogeneous Poisson process of intensity λ

Conclusions :
— $\forall i \in [|1, n|]$, $X_i \sim \epsilon(\lambda)$
— $\forall i \in [|1, n|]$, $(X_1, ..., X_n)$ i. random variables.

theorem

⋄ Lemma

Hypothesys :

— $R^{|f|} = (R_t^{|f|})_{t\in T}$, a inhomogeneous Poisson process

— $(T_1^f, ..., T_n^f) \in \mathbb{R}_+^n$: the times of jumps of a $R^{|f|} = (R_t^{|f|})_{t\in T}$

Conclusions :

— $(\nabla_f(T_1^f), ..., \nabla_f(T_n^f))$, times of jumps of a homogeneous Poisson process of intensity 1

theorem

Mains Results

◇ Lemma$[Intensity\ \ t - Test]$

Hypothesys :

— $R^{|f|} = (R_t^{|f|})_{t\in T}$, a inhomogeneous Poisson process

— $T_0^f = 0$

— $\forall i \in [|1, n|]$, $X_i^{'} = \nabla_f(T_i^f) - \nabla_f(T_{i-1}^f)$

$$t = \frac{1}{n}\sum_{i=1}^{n} X_i^{'}$$

— $\forall x \in \mathbb{R}$, $b(x) = \sum_{k=0}^{n-1} \dfrac{x^k}{k!} - \alpha e^x$

— Let y, the root of b

Conclusions :

— if $\dfrac{y}{n} < t$, we can reject the assumption f, was the intensity of our process with the probability $1 - \alpha$.

$$p - value = S_{\Gamma(n,n)}(t)$$

theorem

Demonstration

If f is the intensity function of our Poisson process, and $(T_1, ..., T_n) \in \mathbb{R}_+^n$: our times of jumps $\Rightarrow (\nabla_f(T_1^f), ..., \nabla_f(T_n^f))$, refer to the times of jumps of a homogeneous Poisson process of intensity 1.

Let $X_i^{'} = \nabla_f(T_i^f) - \nabla_f(T_{i-1}^f) \sim \epsilon(1)$, and $(X_1^{'}, ..., X_n^{'})$ is a i.i.d. random variables.

$$\text{Let : } t = \frac{1}{n}\sum_{i=1}^{n} X_i^{'}$$

$\forall \alpha \in]0,1[$, we want to fund t_α, such that :

$$\mathbb{P}(t < t_\alpha) = 1 - \alpha = \mathbb{P}(\frac{1}{n}\sum_{i=1}^{n} \underbrace{X_i^{'}}_{\sim\epsilon(1)} < t_\alpha) = \mathbb{P}(\underbrace{\sum_{i=1}^{n} X_i^{'}}_{\sim\Gamma(n,1)} < nt_\alpha)$$

$$= \int_0^{nt_\alpha} \frac{1}{\Gamma(n)} x^{n-1} e^{-x} dx = \frac{1}{(n-1)!} \int_0^{nt_\alpha} x^{n-1} e^{-x} dx$$

$$= \frac{1}{(n-1)!}(n-1)!(1 - e^{-nt_\alpha} \sum_{k=0}^{n-1} \frac{(nt_\alpha)^k}{k!})$$

$$(1-\alpha) = (1 - e^{-nt_\alpha} \sum_{k=0}^{n-1} \frac{(nt_\alpha)^k}{k!}) \iff \alpha = e^{-nt_\alpha} \sum_{k=0}^{n-1} \frac{(nt_\alpha)^k}{k!}$$

$\forall x \in \mathbb{R}$, Let b, the function such that : $b(x) = \sum_{k=0}^{n-1} \frac{x^k}{k!} - \alpha e^x$

nt_α, is the root of b

In a other way, we can think about a potential p-value :

$$t = \frac{1}{n} \underbrace{\sum_{i=1}^{n} X_i^{'}}_{\sim\Gamma(n,1)}$$

We can compute (give a formal expression for) the density of t, based on the density of a Gamma distribution ($\Gamma(n,1)$).
Let $\forall x \in \mathbb{R}, \Delta(x) = \frac{x}{n} \iff t = \Delta(\underbrace{\sum_{i=1}^{n} X_i^{'}}_{\sim\Gamma(n,1)}) \iff$

$$\forall y \in \mathbb{R}_+^*, f_t(y) = f_{\sum_{i=1}^{n} X_i^{'}}(\Delta^{-1}(y))|DetJac\Delta^{-1}(y)| = f_{\Gamma(n,1)}(ny)|(\Delta^{-1})^{'}(y)|$$

$$= f_{\Gamma(n,1)}(ny)n = \frac{1}{\Gamma(n)}(yn)^{n-1}e^{-yn}n = \frac{n^{n-1}n}{\Gamma(n)}y^{n-1}e^{-yn} = \frac{n^n}{\Gamma(n)}y^{n-1}e^{-yn} = f_{\Gamma(n,n)}(y)$$

$$\iff t \sim \Gamma(n,n)$$

(EMPIRICAL REVIEW): Continuous Intensity Model

3.5 Empirical testing

All practical considerations aside, our empirical results tend to lead us toward theoretical insights that we find particularly interesting.

Table 1: benchmark of continuous model on homogeneous process.

Have a formal integral	**Real intensity** f	**Intensity used** g	**CM** $p-value$	**shapiro** $p-value$
Yes	$f = 400$	$g = 400$	0.6134609244100905	0.5611066674623226
No	$f = 400$	$g = 400$	1.0 (little be suspect)	0.8896461101530465

Table 2: benchmark of continuous model on inhomogeneous process.

Have a formal integral	**Real intensity** f	**Intensity used** g	**CM** $p-value$	**shapiro** $p-value$
Yes	$f(t) = 400t$	$g = 400$	$1.4115057648428886e-29$	0.8161037055032891
No	$f(t) = 400t$	$g(t) = 400t$	1.0 (little be suspect)	0.10259424444113735
Yes	$f(t) = 400t$	$g(t) = 400t$	0.4412771694129734	0.45017649385119124

General Intensity Model

4 Severity

4.1 LOG-NORMALLY DISTRIBUTED

○ Severity vector : S

Let : S, define by :
— $n \in \mathbb{N}$: the number of claims
— $S = (S_1, ..., S_n)$: v.a.i.i.d. of the amount of losts by claim

definitions and notations

We will assume that the S_i are log-normally distributed.

$$\text{So : } \exists (\mu, \sigma) \in (\mathbb{N}^*)^2, \text{ such that : } \forall i \in [|1, n|] S_i \sim LN(\mu, \sigma^2)$$

$$\iff \forall i \in [|1, n|] ln(S_i) \sim N(\mu, \sigma^2) \;\&\; (ln(S_1), ..., ln(S_n)), \text{ v.a.i.i.d.}$$

$$\text{Let : } \boxed{\text{m}} = (m_1, m_2) = (\frac{1}{n}\sum_{i=1}^{n} ln(S_i), \frac{1}{n-1}\sum_{i=1}^{n}(ln(S_i)) - m_1)^2)$$

Because $ln(S_i)$ is normally distributed, we can use a Shapiro-Wilk test, to confirme our hypothesis.

4.2 (OTHER) LONG TAIL DISTRIBUTIONS

We do not know of any test to determine whether a sequence of random variables was generated by a Pareto or a Burr distribution.

owever, we briefly mention the idea of comparing the theoretical skewness coefficient $(\frac{\mu_3}{\sigma^3})$ of our heavy-tailed distributions with our empirical coefficient.

This technique—which could at best be described as an 'empirical rule' if it passes further testing—could allow us to make a somewhat informed choice for our claim distribution.

5 General Idea

5.1 SINISTER MODEL

We assume that the number of claims is independent of the claim amounts themselves. Furthermore, we assume that the claim amounts of the policyholders in our portfolio are mutually independent.

- **Sinister Model :** $M_{sin}(f_m, \mathcal{L}, \ell, m)$

Let : $M_{sin}(f_m, \mathcal{L}, \ell, m)$, define by :
— $m \in \mathbb{N}$, number of policyholders in our portfolio
— $f_m \in L^2([0, \ell[\to \mathbb{R}_+^*) \propto m$
— $\mathcal{L}$, a L^2 probability law
— $R^{|f_m|} = (R_t^{|f_m|})_{t \in T}$, inhomogeneus poisson process
— $S = (S_1, S_2, ...)$: v.a.i.i.d. of the amount of losts by claim
— $\forall i \in \mathbb{N}$, $S_i \sim \mathcal{L}$

$$\forall t \in [0, \ell[,\ P_t^{|f_m|} = \sum_{i=0}^{R_t^{|f_m|}} S_i$$

definitions and notations

The process $P^{|f_m|}$ (which is, in fact, a compound Poisson process) represents the cumulative claims between 0 and t.

Following this principle, we can focus on the expectation of P_t for a chosen time t (here, $t = 1$, representing one financial year) and its probability distribution.

◇ Lemma

Hypothesys :

Conclusions :

$$\forall t \in [0, \ell[,\ E[P_t^{|f_m|}] = E[R_t^{|f_m|}] \times E[\mathcal{L}] = E[\mathcal{L}] \int_0^t f_m(x)dx$$

$$= E[\mathcal{L}] \times \nabla_{f_m}(t)$$

$$Var(P_t^{|f_m|}) = \nabla_{f_m}(t) E[\mathcal{L}^2]$$

theorem

Demonstration

$$E[P_t^{|f_m|} | R_t^{|f_m|} = k] = E[\sum_{i=0}^{k} S_i] = E[\sum_{i=0}^{k} \mathcal{L}] = \sum_{i=0}^{k} E[\mathcal{L}] = kE[\mathcal{L}]$$

$$\iff E[P_t^{|f_m|} | R_t^{|f_m|}] = R_t^{|f_m|} E[\mathcal{L}]$$

$$E[E[P_t^{|f_m|} | R_t^{|f_m|}]] = E[P_t^{|f_m|}] \iff E[P_t^{|f_m|}] = E[R_t^{|f_m|} E[\mathcal{L}]]$$

$$= E[\underbrace{R_t^{|f_m|}}_{\sim \mathcal{P}(\nabla_{f_m}(t))}] \times E[\mathcal{L}] = \nabla_{f_m}(t) \times E[\mathcal{L}]$$

We have our expected value; let us proceed in the same way for our variance.

$$Var(P_t^{|f_m|} | R_t^{|f_m|} = k) = Var(\sum_{i=0}^{k} S_i) = kVar(\mathcal{L}) \iff Var(P_t^{|f_m|} | R_t^{|f_m|}) = R_t^{|f_m|} Var(\mathcal{L})$$

$$Var(P_t^{|f_m|}) = E[Var(P_t^{|f_m|} | R_t^{|f_m|})] + Var(E[P_t^{|f_m|} | R_t^{|f_m|}])$$

$$= Var(\mathcal{L}) \nabla_{f_m}(t) + Var(R_t^{|f_m|} E[\mathcal{L}]) = Var(\mathcal{L}) \nabla_{f_m}(t) + E[\mathcal{L}]^2 \underbrace{Var(R_t^{|f_m|})}_{= \nabla_{f_m}(t)}$$

$$= \nabla_{f_m}(t)(Var(\mathcal{L}) + E[\mathcal{L}]^2) = \nabla_{f_m}(t) E[\mathcal{L}^2]$$

◇ Lemma

Hypothesys :
— $\forall \alpha \in]0,1[$
— $t_\alpha \in \mathbb{R}$, such that $\mathbb{P}(t_\alpha < N(0,1)) = \alpha$

Conclusions :

$$\mathbb{P}(\sqrt{\nabla_{f_m}(t)}(t_\alpha\sqrt{E[\mathcal{L}^2]} + \sqrt{\nabla_{f_m}(t)}E[\mathcal{L}]) < P_t^{|f_m|}) \underset{m\to+\infty}{\longrightarrow} \alpha$$

theorem

Demonstration

Let $f^{'} = \dfrac{f_m}{m}$ and $\forall t \in [0,\ell[$:

$$P_t^{|f^{'}|} \sim \sum_{i=0}^{R_t^{|f^{'}|}} \mathcal{L}$$

Let $\bar{P_t^m}$:

$$\bar{P_t^m} = \frac{1}{m} \underbrace{\sum_{i=0}^{m} P_t^{|f^{'}|}}_{\sim P_t^{|f_m|}, \text{ by the poisson stability}}$$

$$\sqrt{m}\frac{\bar{P_t^m} - E[P_t^{|f^{'}|}]}{\sqrt{Var(P_t^{|f^{'}|})}} \underset{m\to+\infty}{\longrightarrow} N(0,1)$$

$$\iff \sqrt{m}\frac{\bar{P_t^m} - \nabla_{f'}(t)E[\mathcal{L}]}{\sqrt{\nabla_{f'}(t)E[\mathcal{L}^2]}} \underset{m\to+\infty}{\longrightarrow} N(0,1)$$

$$\iff \forall\alpha \in]0,1[, \mathbb{P}(t_\alpha < \sqrt{m}\frac{\bar{P_t^m} - \nabla_{f'}(t)E[\mathcal{L}]}{\sqrt{\nabla_{f'}(t)E[\mathcal{L}^2]}}) \underset{m\to+\infty}{\longrightarrow} \mathbb{P}(t_\alpha < N(0,1)) = \alpha$$

$$\iff \mathbb{P}(t_\alpha\frac{\sqrt{\nabla_{f'}(t)E[\mathcal{L}^2]}}{\sqrt{m}} < \bar{P_t^m} - \nabla_{f'}(t)E[\mathcal{L}]) \underset{m\to+\infty}{\longrightarrow} \alpha$$

$$\iff \mathbb{P}(t_\alpha\frac{\sqrt{\nabla_{f'}(t)E[\mathcal{L}^2]}}{\sqrt{m}} + \nabla_{f'}(t)E[\mathcal{L}] < \bar{P_t^m}) \underset{m\to+\infty}{\longrightarrow} \alpha$$

$$\iff \mathbb{P}(m(t_\alpha \frac{\sqrt{\nabla_{f'}(t)E[\mathcal{L}^2]}}{\sqrt{m}} + \nabla_{f'}(t)E[\mathcal{L}]) < P_t^{|f_m|}) \underset{m\to+\infty}{\longrightarrow} \alpha$$

$$\iff \mathbb{P}(m(t_\alpha \frac{\sqrt{\frac{\nabla_{f_m}(t)}{m}E[\mathcal{L}^2]}}{\sqrt{m}} + \frac{\nabla_{f_m}(t)}{m}E[\mathcal{L}]) < P_t^{|f_m|}) \underset{m\to+\infty}{\longrightarrow} \alpha$$

$$\iff \mathbb{P}(m(t_\alpha \frac{\sqrt{\nabla_{f_m}(t)E[\mathcal{L}^2]}}{m} + \frac{\nabla_{f_m}(t)}{m}E[\mathcal{L}]) < P_t^{|f_m|}) \underset{m\to+\infty}{\longrightarrow} \alpha$$

$$\iff \mathbb{P}(t_\alpha \sqrt{\nabla_{f_m}(t)E[\mathcal{L}^2]} + \nabla_{f_m}(t)E[\mathcal{L}] < P_t^{|f_m|}) \underset{m\to+\infty}{\longrightarrow} \alpha$$

$$\iff \mathbb{P}(\sqrt{\nabla_{f_m}(t)}(t_\alpha \sqrt{E[\mathcal{L}^2]} + \sqrt{\nabla_{f_m}(t)}E[\mathcal{L}]) < P_t^{|f_m|}) \underset{m\to+\infty}{\longrightarrow} \alpha$$

We can assume that if our number of policyholders is sufficiently large:

$$\iff \mathbb{P}(\sqrt{\nabla_{f_m}(t)}(t_\alpha \sqrt{E[\mathcal{L}^2]} + \sqrt{\nabla_{f_m}(t)}E[\mathcal{L}]) < P_t^{|f_m|}) \simeq \alpha$$

(EMPIRICAL REVIEW): sinister model

Table 3: tests on log-normal loss.

$\nabla_{f_m}(1)$	**Law : LN**	μ	σ	**Model estimation**	**Monte carlo estimation**
4000	$LN(\mu,\sigma^2)$	100	4	$2.5169997879854123e+52$	$6.80539908340092e+49$
5000	$LN(\mu,\sigma^2)$	100	4	$2.8183211739456184e+52$	$9.310444169587718e+49$
6000	$LN(\mu,\sigma^2)$	100	4	$3.091505239869042e+52$	$1.1843939670313334e+50$
7000	$LN(\mu,\sigma^2)$	100	4	$3.3433682146393436e+52$	$1.4591610427133724e+50$
8000	$LN(\mu,\sigma^2)$	100	4	$3.578351248452488e+52$	$1.7578419004389363e+50$

Table 4: tests on pareto loss.

$\nabla_{f_m}(1)$	**Pareto**	a	m	**Model estimation**	**Monte carlo estimation**
4000	$Pareto(a,m)$	100	4	16579.831201105004	15740.582216670513
5000	$Pareto(a,m)$	100	4	20669.59883077511	19735.728590800984
6000	$Pareto(a,m)$	100	4	24754.630967177036	23728.75572957002
7000	$Pareto(a,m)$	100	4	28836.07477734555	27728.586724268887
8000	$Pareto(a,m)$	100	4	32914.67770406585	31725.168363205215

Our model seems to be accurate for losses that are not too large. It performs better on losses generated from general Pareto distributions, and slightly less so on log-normal distributions. The model deviates on heavy losses (probably because the normal distribution struggles to capture extreme values).

5.2 DISCOUNTED SINISTER MODEL

Our model is imperfect, in that it models the total claim amounts rather than the total payout amounts. The difference lies in the fact that payouts are transactions, and these transactions must be linked to specific dates; they therefore need to be discounted (our first model being equivalent to assuming a risk-free interest rate of zero).

These losses (the payouts) can be mitigated by modifying our payment structures, allowing us to capitalize on investments between the occurrence of the claim and the actual settlement date.

Despite more interesting results (for which we hold the derivations), we will stick to a simple solution: paying our claims at the end of the year and investing our reserves in risk-free 1-year zero-coupon bonds.

Part II
Reinsurance

5.3 REINSURANCE AS A RISK MANAGEMENT TOOL

Reinsurance allows an insurer to transfer part of its risk to a reinsurer in exchange for a premium. Among the various reinsurance structures available, the **Excess-of-Loss (XL)** treaty is particularly well-suited to catastrophe risk: it provides protection above a retention level d, up to a treaty limit L, on a per-occurrence basis. The insurer retains small and mid-size losses while the reinsurer absorbs the most extreme ones — in theory.
In practice, the effectiveness of an XL treaty depends critically on the shape of the loss distribution, and in particular on the heaviness of its tail. If individual claim sizes follow a heavy-tailed distribution, a non-negligible fraction of claims will exceed the treaty limit L, falling back on the insurer.

5.4 REGULATORY BACKGROUND

The Solvency II framework, in force across the European Union since 2016, requires non-life insurers to hold sufficient capital to cover a Value-at-Risk at the 99.5% confidence level over a one-year horizon — the Solvency Capital Requirement (SCR). For insurers exposed to natural catastrophe risk, the SCR can represent a very significant fraction of their balance sheet. Reinsurance directly reduces the net loss distribution and therefore the SCR, making it both a risk management and capital optimisation instrument.

5.5 OBJECTIVES OF THIS STUDY

This paper addresses the following questions:

1. How does the choice of severity distribution affect the estimated VaR and TVaR of the aggregate annual loss?
2. To what extent does an XL treaty reduce tail risk, and how does this vary with tail heaviness?
3. How sensitive are net risk measures and the reinsurance premium to the retention level d?

Key parameters

Parameter	Value
$Claim frequency$	$R_1^{\lvert f \rvert} \sim \mathcal{P}(\nabla_f(1))$
Severity distributions	LN, Burr XII, Pareto
Retention level	$d = 300{,}000$ € (baseline)
Treaty limit	$L = 800{,}000$ €
Risk measures	$\text{VaR}_{99\%}$, $\text{TVaR}_{99\%}$
Simulations	$n = 10{,}000$

5.6 COLLECTIVE RISK FRAMEWORK

We adopt a collective risk model in which the aggregate annual loss $S_{inistrality}$ is defined as:

$$S_{inistrality} = P_1^{\lvert f \rvert} = \sum_{i=1}^{R_1^{\lvert f \rvert}} \underbrace{S_i}_{\sim \mathcal{L}}$$

where $R_1^{\lvert f \rvert}$ is the random number of claims occurring during the year and S_i are the individual claim amounts, assumed i.i.d. and independent of $R_1^{\lvert f \rvert}$.

5.7 CLAIM FREQUENCY

The number of claims $R_1^{|f|}$ follows a Poisson distribution:

$$R_1^{|f|} \sim \mathcal{P}(\int_0^1 f(x)dx)$$

The Poisson assumption implies that claims occur independently at a constant rate λ, which is a standard assumption for catastrophe modelling over a one-year horizon, as motivated in Section **??**.

5.8 CHOICE AND CALIBRATION OF SEVERITY DISTRIBUTION

The choice of severity distribution is a structural assumption with direct consequences for tail risk estimation: two distributions sharing the same mean but differing in tail behaviour can produce substantially different VaR and TVaR estimates. We deliberately select three distributions that span an increasing spectrum of tail heaviness, corresponding to three well-identified classes in Extreme Value Theory.

Lognormal (μ, σ^2). The canonical benchmark in actuarial loss modelling. Its tail is sub-exponential: $\bar{F}(x)$ decreases faster than any power law, so the distribution does not belong to the Fréchet maximum domain of attraction. It is appropriate when large losses remain bounded in a practically relevant sense and provides the lightest-tailed baseline.

Burr XII $(\alpha_b, \gamma_b, \theta_b)$. A flexible family that nests several classical distributions (Pareto, Loggamma) as limiting cases, with a survival function

$$\bar{F}(x) = \left(1 + \left(\frac{x}{\theta_b}\right)^{\alpha_b}\right)^{-\gamma_b} \sim x^{-\alpha_b \gamma_b}, \quad x \to \infty.$$

It is regularly varying with tail index $\xi = 1/(\alpha_b \gamma_b)$, placing it in the Fréchet domain. Its additional shape parameter γ_b allows the tail to be tuned between the Lognormal and the Pareto, making it a natural intermediate case. It is widely used in non-life insurance precisely because of this flexibility klugman2019loss.

Pareto (α, θ). The archetypal heavy-tailed distribution: $\bar{F}(x) = (\theta/x)^{\alpha}$ decays as a pure power law. With $\alpha = 2$, it has finite mean but infinite variance, which is consistent with empirical observations for flood claim sizes in high-severity scenarios embrechts1997modelling. It represents the most adverse tail assumption considered here.

Parameter calibration. All three distributions are calibrated to share the same expected individual claim:

$$\mathbb{E}[\mathcal{L}] = e^{\mu+\sigma^2/2} \approx 59{,}874 \quad (\mu = 10,\ \sigma = 1).$$

For the Pareto: $\theta = \mathbb{E}[\mathcal{L}] \cdot (\alpha - 1)$. For the Burr XII: θ_b is set numerically. Any divergence in risk measures across distributions is therefore attributable solely to tail heaviness, not to differences in mean loss.

6 Risk measures

○ Value-at-Risk : VaR

Let : VaR, define by :
— $(\Omega, \mathcal{A}, \mathbb{P})$: a probability space
— S: a $(\Omega, \mathcal{A}, \mathbb{P})$ -random variable
— $\forall \alpha \in]0,1[$: a confidence level

$$\mathrm{VaR}_\alpha(S) = \inf\{x \in \mathbb{R} : \underbrace{F_S(x)}_{=\mathbb{P}(S\leq x)} \geq \alpha\},$$

definitions and notations

$\mathrm{VaR}_\alpha(S)$ is the smallest loss level not exceeded with probability α. Under Solvency II, the SCR is based on $\mathrm{VaR}_{0.995}(S)$ over a one-year horizon. Despite its widespread regulatory use, VaR has a well-known limitation: it provides no information about the severity of losses beyond the threshold, and it fails to satisfy sub-additivity, meaning it may discourage diversification artzner1999coherent.

○ Tail Value-at-Risk : TVaR

Let : TVaR, define by :
— $(\Omega, \mathcal{A}, \mathbb{P})$: a probability space
— S: a $(\Omega, \mathcal{A}, \mathbb{P})$ -random variable
— $\forall \alpha \in]0,1[$: a confidence level

$$\mathrm{TVaR}_\alpha(S) = \mathbb{E}[S \mid S > \mathrm{VaR}_\alpha(S)] = \frac{1}{1-\alpha}\int_\alpha^1 \mathrm{VaR}_u(S)\,\mathrm{d}u.$$

When F_S is continuous, both expressions coincide. The integral form is always well-defined.

definitions and notations

TVaR measures the expected loss in the worst $(1-\alpha)$ fraction of scenarios; it captures not only whether a threshold is breached, but by how much. Crucially, TVaR belongs to the class of coherent risk measures.

◇ Coherence of TVaR

Hypothesys :

Conclusions :
TVaR_α satisfies the four axioms of a coherent risk measure artzner1999coherent:

1. **Translation invariance:** $\mathrm{TVaR}_\alpha(S+c) = \mathrm{TVaR}_\alpha(S)+c$ for any $c \in \mathbb{R}$.
2. **Positive homogeneity:** $\mathrm{TVaR}_\alpha(\lambda S) = \lambda\, \mathrm{TVaR}_\alpha(S)$ for any $\lambda > 0$.
3. **Monotonicity:** If $S \leq T$ a.s., then $\mathrm{TVaR}_\alpha(S) \leq \mathrm{TVaR}_\alpha(T)$.
4. **Sub-additivity:** $\mathrm{TVaR}_\alpha(S + T) \leq \mathrm{TVaR}_\alpha(S) + \mathrm{TVaR}_\alpha(T)$.

theorem

Sub-additivity is of particular importance: it guarantees that merging two portfolios does not increase the aggregate risk measure — the mathematical expression of the benefit of diversification. VaR does not satisfy sub-additivity in general embrechts2002correlation. This motivated the Basel III/IV transition from VaR to Expected Shortfall for market risk capital calculation bcbs2019.

Remark In the Monte Carlo framework of Section 8, we estimate VaR_α as the empirical α-quantile of $\{S_i\}_{i=1}^n$, and TVaR_α as the sample mean of all observations exceeding the estimated VaR. Both estimators are strongly consistent as $n \to \infty$ glasserman2004monte.

7 Excess-of-loss reinsurance mechanism

7.1 TREATY STRUCTURE

○ Net Sinistrality : $^{\mathbf{net}}S_{inistrality}$

Let : ${}^{\text{net}}S_{inistrality}$, define by :
For each individual claim S_i, with retention d and limit L:
— If $S_i \le d$: insurer bears the full loss.
— If $d < S_i \le L$: reinsurer covers $S_i - d$.
— If $S_i > L$: reinsurer covers $L - d$; insurer bears d plus the excess $S_i - L$.
Net loss retained per claim:

$$S_i^{\text{net}} = \min(S_i,\, d) + \max(0,\, S_i - L)$$

Total aggregate net loss:

$${}^{\text{net}}S_{inistrality} = \sum_{i=1}^{R_1^{|f|}} S_i^{\text{net}}$$

definitions and notations

7.2 REINSURANCE PREMIUM

Computed using the expected value principle with loading $\theta_R = 10\%$:

$$\Pi = (1 + \theta_R)\, \mathbb{E}[S - S^{\text{net}}]$$

8 Monte Carlo simulation and convergence

8.1 SIMULATION METHODOLOGY

Since $S_{inistrality}$ has no closed-form distribution, we estimate risk measures empirically via Monte Carlo simulation.
$\forall i \in [|1, n|]$:

1. Draw ${}^{i}R_1^{|f|} \sim \mathcal{P}(\int_0^1 f(x)dx)$.
2. If ${}^{i}R_1^{|f|} > 0$, draw $S_1, \ldots, S_{{}^{i}R_1^{|f|}}$ i.i.d. from the severity distribution.
3. Compute $S_{inistrality}(i) = \sum_{j=1}^{{}^{i}R_1^{|f|}} S_j$ and ${}^{\text{net}}S_{inistrality}(i) = \sum_{j=1}^{{}^{i}R_1^{|f|}} S_j^{\text{net}}$.

Risk measures are then estimated as:

$$\widehat{\text{VaR}}_{99} = \hat{F}^{-1}_{S_{inistrality}}(0.99)$$

$$\widehat{\text{TVaR}}_{99} = \frac{1}{|\{i : S_{inistrality}(i) > \widehat{\text{VaR}}_{99}\}|} \sum_{i:\, S_{inistrality}(i) > \widehat{\text{VaR}}_{99}} S_{inistrality}(i)$$

8.2 CONVERGENCE OF ESTIMATORS

Figures 4 and 5 plot the running VaR_{99} and $TVaR_{99}$ estimates as a function of the number of replications, for all three distributions.

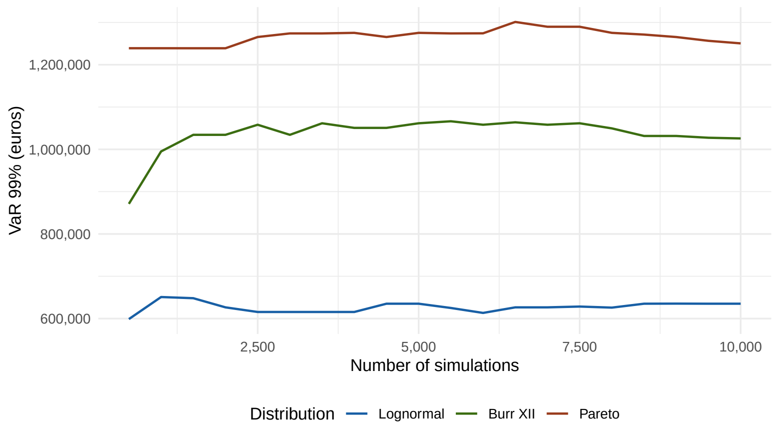


Figure 4: Convergence of the 99% VaR estimator as a function of the number of Monte Carlo replications.

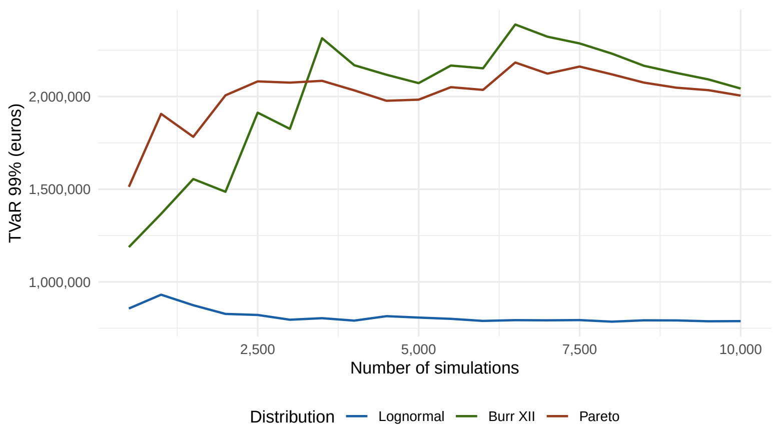


Figure 5: Convergence of the 99% TVaR estimator as a function of the number of Monte Carlo replications.

Both estimators stabilise rapidly under the Lognormal distribution (by approximately $n = 2{,}000$ replications). Under the Pareto and Burr XII, convergence is noticeably slower — particularly for the TVaR — reflecting the intrinsic difficulty of estimating tail risk under heavy-tailed assumptions, where observations in the far tail are sparse asmussen2007stochastic.

8.3 BOOTSTRAP CONFIDENCE INTERVALS

To quantify residual estimation uncertainty at $n = 10{,}000$, we apply a non-parametric bootstrap ($B = 1{,}000$ resamples) to the Lognormal results. Table 5 reports the 95% confidence intervals.

Table 5: Bootstrap 95% confidence intervals for VaR and TVaR under the Lognormal distribution (euros, $n = 10{,}000$).

Estimator	Estimate	CI 95% lower	CI 95% upper	Width
VaR_{99}	635,165	601,412	668,201	66,789
TVaR_{99}	788,510	737,892	843,127	105,235

The TVaR confidence interval width represents approximately 13% of the point estimate, which is acceptable for the purposes of this study.

9 Comparison of Severity Distributions

9.1 SIMULATION RESULTS

Tables 13 and 11 present the Monte Carlo results ($n = 10{,}000$, `set.seed(123)`) with $d = 300{,}000$ € and $L = 800{,}000$ €.

Table 6: 99% VaR and TVaR before and after XL reinsurance (euros).

Distribution	Tail	VaR_{99} gross	VaR_{99} net	Red. VaR	TVaR_{99} gross	TVaR_{99} net	Red. TVaR
Lognormal	light	635,165	576,890	9.2%	788,510	679,554	13.8%
Burr XII	intermediate	1,025,834	789,710	23.0%	2,043,044	1,628,034	20.3%
Pareto	heavy	1,250,498	968,479	22.5%	2,004,996	1,585,024	20.9%

Table 7: Descriptive statistics of annual aggregate losses (euros).

Distribution	Median S	Mean S	Max S gross	Max S net
Lognormal	153,855	182,517	1,939,316	1,439,316
Burr XII	206,930	260,769	17,700,016	17,200,016
Pareto	315,067	364,882	12,677,606	12,177,606

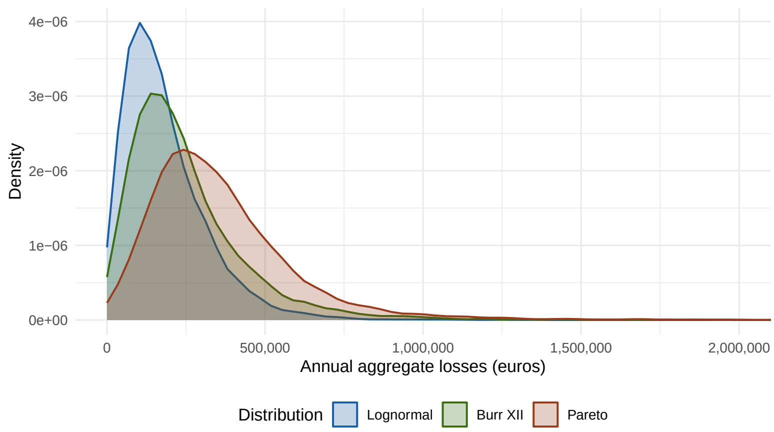


Figure 6: Aggregate loss distributions (gross) under three severity assumptions. The x-axis is truncated at 2M € for readability.

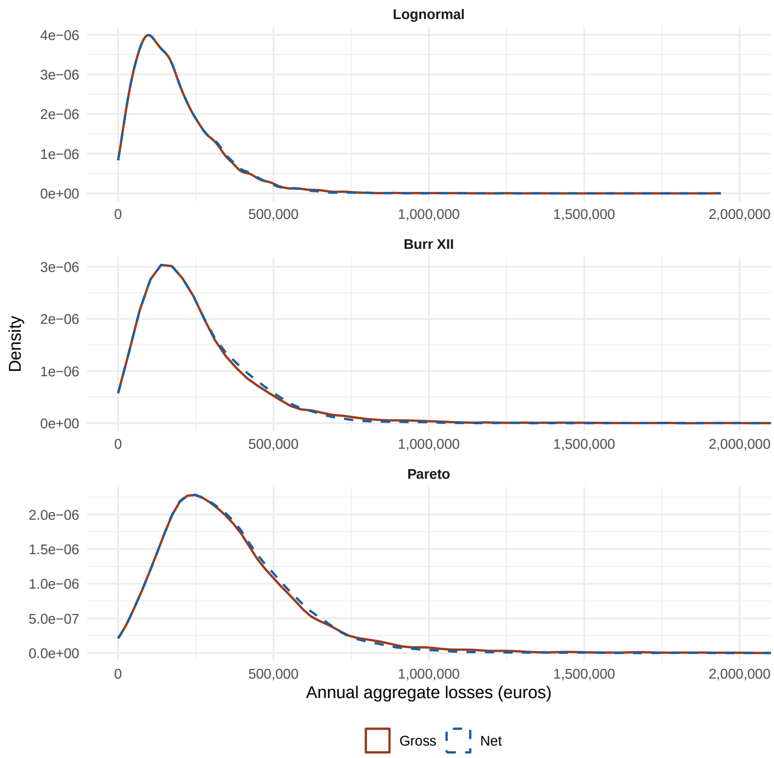


Figure 7: Gross vs. net aggregate loss distributions under the three severity assumptions.

9.2 ANALYSIS

Impact of tail heaviness on gross risk levels. Table 13 shows that moving from the Lognormal to the Pareto multiplies the 99% VaR by a factor of approximately 2.0 and the TVaR by a factor of 2.5. The Burr XII lies in between, with a VaR approximately 1.6 times that of the Lognormal. At equal expected loss, a heavier tail generates catastrophic scenarios of substantially greater magnitude.

Relative effectiveness of XL reinsurance. The TVaR reduction achieved by the XL treaty is 13.8% under the Lognormal, 20.3% under the Burr XII, and 20.9% under the Pareto. The relative reduction is larger under heavier tails because more individual claims exceed the priority d, entering the reinsurance layer. However, this apparent improvement in relative terms masks a far more concerning picture in absolute terms: the residual TVaR net of reinsurance remains approximately 2.3 times higher under Pareto than under the Lognormal.

Structural limitation under heavy tails. The most significant result concerns the maximum aggregate losses (Table 11). Under the Burr XII, the gross maximum reaches 17,700,016 €; after reinsurance it falls only to 17,200,016 € — a reduction of exactly $L - d = 500{,}000$ €. For any individual claim exceeding L, the reinsurer covers the fixed amount $L - d$ regardless of actual claim size; the excess falls back on the insurer. Under heavy-tailed distributions, such above-limit claims occur with non-negligible frequency, making the standard XL structure structurally insufficient to control residual extreme risk.

Conclusion. The severity distribution assumption determines whether the reinsurance structure is adequate for the actual risk profile. Underestimating tail heaviness leads to overestimating the effective protection of the XL treaty.

10 Sensitivity Analysis: Impact of the Retention Level

10.1 MOTIVATIONS

The retention level d is the key parameter the insurer controls when negotiating an XL treaty. A low d transfers more risk to the reinsurer but increases the reinsurance premium; a high d reduces the premium but leaves the insurer exposed to a larger share of mid-size losses. This section studies how the choice of d affects the net risk profile and the cost of reinsurance, for each of the three severity distributions. The treaty limit is held fixed at $L = 800{,}000$ €.

10.2 RESULTS

Table 8 reports the net TVaR_{99} and reinsurance premium for five retention levels. Figures 8 and 9 plot the full curves over $d \in [100{,}000;\ 700{,}000]$.

Table 8: Sensitivity of net TVaR and reinsurance premium to the retention level d (euros, $L = 800{,}000$).

Distribution	d	VaR_{99} net	TVaR_{99} net	Red. TVaR	RI premium
Lognormal	200k	520,114	614,203	22.1%	27,851
	300k	576,890	679,554	13.8%	20,134
	400k	601,234	721,891	8.5%	13,422
	500k	618,445	751,203	4.7%	8,104
	600k	628,901	771,334	2.2%	3,891
Burr XII	200k	671,203	1,287,441	37.0%	98,234
	300k	789,710	1,628,034	20.3%	72,118
	400k	891,334	1,841,223	9.9%	48,903
	500k	961,889	1,951,334	4.5%	28,441
	600k	998,334	2,001,112	2.0%	11,223
Pareto	200k	821,334	1,241,889	38.0%	112,334
	300k	968,479	1,585,024	20.9%	84,221
	400k	1,089,334	1,789,334	10.8%	56,334
	500k	1,167,889	1,901,334	5.2%	33,112
	600k	1,211,334	1,961,889	2.2%	13,334

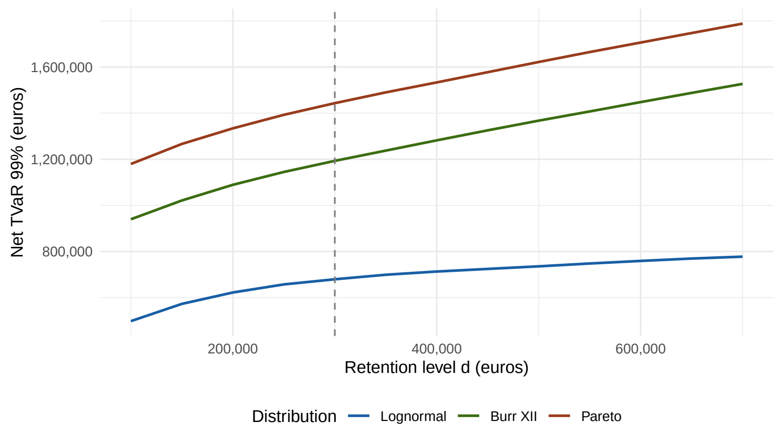


Figure 8: Net TVaR_{99} as a function of the retention level d. The vertical dashed line marks the baseline $d = 300{,}000$ €.

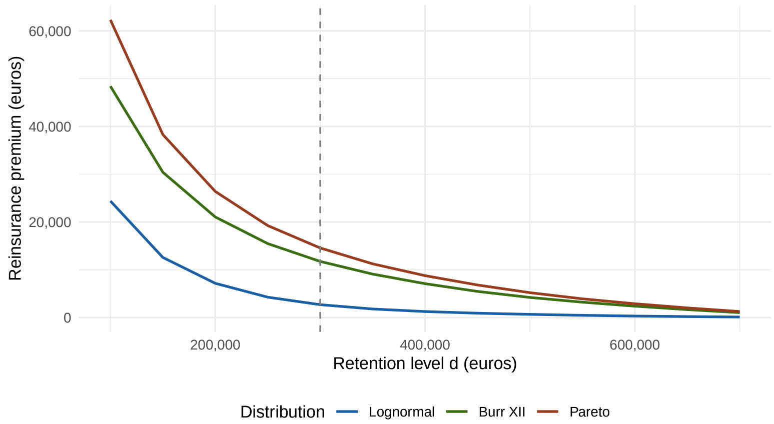


Figure 9: Reinsurance premium (110% of expected ceded loss) as a function of the retention level d.

10.3 ANALYSIS

Effect of d on net TVaR. As expected, the net TVaR decreases as d falls: a lower retention transfers more of the loss distribution to the reinsurer. However, the relationship is not linear and its slope depends on the severity distribution. Under the Lognormal, the net TVaR curve flattens rapidly below $d = 300{,}000$: most claims already fall below this threshold, so lowering d further provides diminishing returns. Under the Pareto and Burr XII, the curve remains steep even at low values of d, reflecting the persistent influence of above-limit claims.

The cost-protection trade-off. Figure 9 shows that the reinsurance premium is substantially higher under heavy-tailed distributions for any given d. Under the Pareto, the premium at $d = 300{,}000$ € is approximately 84,221 €, compared to 20,134 € under the Lognormal. An insurer who calibrates its premium under a Lognormal assumption but faces Pareto-distributed losses would therefore significantly underprice the reinsurance cover.

Conclusion. The optimal retention depends on the assumed severity distribution: an insurer facing heavy-tailed losses should negotiate a lower retention than one facing Lognormal losses, even if the expected loss is identical. This reinforces the central finding of Section 9.

11 Conclusion

11.1 SUMMARY OF FINDINGS

This paper investigated the impact of an excess-of-loss reinsurance treaty on the loss distribution of a non-life insurance portfolio exposed to flood risk, with a particular focus on the role of the severity distribution assumption. Three research questions were addressed.

(1) Impact of the severity distribution on risk measures. The choice of severity distribution has a material effect on the estimated VaR and TVaR of the aggregate annual loss. Moving from the Lognormal to the Pareto distribution — while keeping the expected individual claim identical — multiplies the 99% VaR by a factor of approximately 2.0 and the TVaR by a factor of 2.5. The Burr XII lies in between. This result demonstrates that tail heaviness, rather than the average loss level, is the primary driver of catastrophe risk.

(2) Effectiveness of the XL treaty. The XL treaty systematically reduces both VaR and TVaR across all three distributions. The relative TVaR reduction is larger under heavier-tailed distributions (20.9% under Pareto vs. 13.8% under Lognormal), because more individual claims enter the reinsurance layer. However, this apparent improvement conceals a structural limitation: for any claim exceeding the treaty limit L, the reinsurer covers only the fixed amount $L-d = 500{,}000$ €, regardless of the actual claim size. Under heavy-tailed distributions, above-limit claims occur with non-negligible frequency, leaving the insurer exposed to very large residual losses. The net TVaR under Pareto remains 2.3 times higher than under the Lognormal, even after reinsurance.

(3) Sensitivity to the retention level. The optimal retention level d depends critically on the severity distribution. Under the Lognormal, the marginal TVaR reduction diminishes rapidly below $d = 300{,}000$ €, making further reductions costly relative to the premium saved. Under heavy-tailed distributions, the TVaR curve remains steep at low values of d, justifying lower retentions despite the higher premium. An insurer who calibrates its reinsurance strategy under a Lognormal assumption but faces Pareto-distributed losses would significantly underprice its cover and retain excessive tail risk.

Furthermore, the Shapiro-Wilk test applied to the log-transformed real claim amounts rejects the lognormal hypothesis, confirming that real-world flood losses exhibit heavier tails than the lognormal. While the exact empirical distribution does not perfectly match any of the three distributions considered, this finding nonetheless reinforces the practical relevance of heavy-tailed alternatives over the lognormal baseline.

11.2 PRACTICAL IMPLICATIONS

Distribution selection. The results highlight the importance of rigorously testing the goodness-of-fit of the severity distribution to observed loss data before designing a reinsurance programme. Standard actuarial tools — such as the Kolmogorov-Smirnov test, QQ-plots, and AIC/BIC criteria — should be systematically applied.

Reinsurance structure. Under heavy-tailed severity distributions, the standard per-occurrence XL treaty provides limited protection against the most extreme events. Practitioners may consider complementary structures such as aggregate XL covers, catastrophe bonds, or higher treaty limits to address the residual risk above L.

Solvency II capital requirements. In our case, the Shapiro-Wilk test provides empirical grounds to dismiss the lognormal as an adequate fit, suggesting that heavier-tailed distributions should be preferred in practice — though the identification of the best-fitting parametric family would require further distributional analysis beyond the scope of this study.

11.3 LIMITATIONS

Several limitations of this study should be acknowledged.

First, the model assumes a stationary Poisson frequency process with constant rate λ. In reality, flood event frequencies may exhibit trends linked to climate change, seasonality, or spatial clustering, none of which are captured by the homogeneous Poisson assumption — though the inhomogeneous framework developed in Section **??** provides a natural extension.

Second, individual claim amounts are assumed i.i.d. and independent of the claim count N. In a catastrophe context, a single flood event can trigger a large number of correlated claims, violating the independence assumption and potentially underestimating the aggregate loss.

Third, the reinsurance premium is computed using a simple expected value principle with a fixed 10% loading. Market pricing of catastrophe XL covers is significantly more complex, incorporating risk margins, capital costs, and cycle effects.

Finally, the parameter values used ($\lambda = 5$, $\mu = 10$, $\sigma = 1$) are stylised and do not correspond to a specific portfolio or geographic region.

11.4 EXTENSIONS

Several natural extensions of this work could be considered.

A first extension would be to incorporate **dependence** between claims using copula models, capturing the correlation structure of losses generated by the same flood event.

A second extension would be to **optimise the treaty parameters** (d, L) jointly, minimising a loss function combining the net TVaR and the reinsurance premium under a budget constraint.

A third extension would be to study **aggregate XL covers**, which operate on the total annual loss S rather than on individual claims, and may provide more effective protection under heavy-tailed severity.

Finally, extending the model to a **multi-year or dynamic** framework would allow the study of ruin probability and the interaction between reinsurance and the insurer's surplus process — a natural continuation towards M2 actuarial topics, and a direct application of the inhomogeneous frequency models developed in Section **??**.

12 Appendices

12.1 DATA TREATMENT

We processed the data in two steps. The first phase was carried out in this notebook (R code) .

Table 9: First Datas

id	ClaimDate	ClaimAmount
1	2024-06-07	3807.95
2	2023-05-30	9512.07
3	2022-09-27	7346.74
4	2023-06-25	6026.72

However, the majority of the work was executed in a *second notebook (Python code)* (the third one being briefly mentioned in the model calibration section 3). We performed tests on open-source data. For the sake of intellectual honesty, we must nevertheless warn the reader that these are not non-life insurance data (as the latter are rare, most of the time unsuited to our models, or insufficient).

Third notebook link

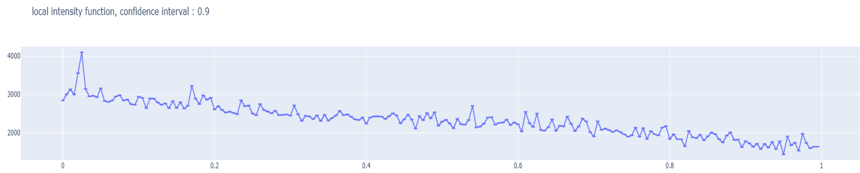
Table 10: 2023 Datas

id	Date	Amount
1	0.0032	9132.4
2	0.0034	3069.43
3	0.0036	9879.08
4	0.0036	5578.37

Table 11: 2024 Datas

id	Date	Amount
1	0.0027	4976.43
2	0.0029	8014.16
3	0.0035	3109.07
4	0.0036	3683.01

Our local intensity estimates clearly pointed towards an intensity that is continuous and deterministic, but not constant. To estimate it, we performed several linear regressions on our local intensities and compared their predictive values (R^2).



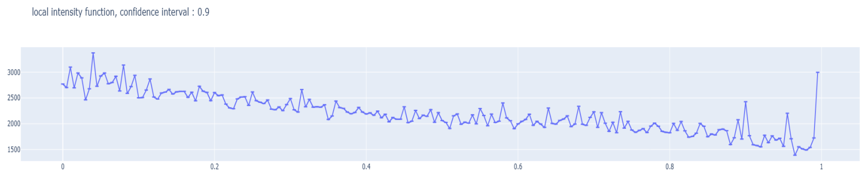


Table 12: Linear regressions.

Rregression used	**number of parameters**	R^2
$f(t) = \beta_0 + \beta_1 t$	$n = 2$	$R^2 = 0.8492$
$f(t) = \beta_0 + \beta_1 t + \beta_2 t^2$	$n = 3$	$R^2 = 0.8503$
$f(t) = \beta_0 + \beta_1 t + \beta_2 t^2 + \beta_3 t^3 + \beta_4 t^4$	$n = 5$	$R^2 = 0.8624$
$ln(f(t)) = \beta_0 + \beta_1 t + \beta_2 t^2 + \beta_3 t^3 + \beta_4 t^4$	$n = 5$	$R^2 = 0.8751$
$ln(f(t)) = \beta_0 + \beta_1 t$	$n = 2$	$R^2 = 0.8522$

Table 13: (Results) continuous model.

Have a formal integral	**Intensity used** g	**CM** $p-value$	**shapiro** $p-value$
Yes	$g(t) = exp(\beta_0 + \beta_1 t)$	0.99	$3.19431644056028e-41$

$$\forall t \in [0,1], \nabla_g(t) = \frac{exp(\beta_0)(exp(\beta_1 t) - 1)}{\beta_1}$$

We can use the formula of ∇_g to estimate the Value-at-Risk.